

Experiment 6: Functions

Objective: To understand the concepts of user-defined functions, and recursion.

Questions:

1. Write a function to find the factorial of a number using recursion.

Sample Input:

Enter Number: 5

Sample Output:

Factorial of 5 is 120

2. Write a function that reverses a given number.

Sample Input:

Enter Number: 1234

Sample Output:

Reversed Number: 4321

3. Write a recursive function to find the sum of first n natural numbers.

Sample Input:

Enter a number: 5

Sample Output:

Sum of first 5 natural numbers = 15

4. Write a recursive function to find GCD using Euclid's algorithm.

Euclid's algorithm is based on this rule:

$$\text{GCD}(a, b) = \text{GCD}(b, a \% b)$$

Repeat until the remainder becomes 0.

Base Case: If $b = 0$ - $\text{GCD} = a$

Sample Input:

Enter two numbers: 48 18

Sample Output:

GCD of 48 and 18 = 6

5. Write a recursive function to convert decimal number to binary.

Sample Input:

Enter a decimal number: 10

Sample Output:

Binary: 1010

6. Given a number n, print triangular pattern. You are allowed to use only one loop.

Sample Input:

Enter n: 4

Sample Output:

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7. Write a recursive function to calculate: $x^n = x \times x \times x \dots (n \text{ times})$

Sample Input:

Enter base: 2

Enter exponent: 3

Sample Output:

2^3 = 8

8. Write a program to find the nth term of the Fibonacci series.

Sample Input:

Enter the value of n: 6

Sample Output:

The Fibonacci term for n = 6 is: 8